首都师范大学
CAPITAL NORMAL UNIVERSITY

电路分析

电阻电路的一般分析方法

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一 → 2b法（2b method）

2b法

支路 $b=6$ b 个支路电压 $2b$ 个变量

b 个支路电流

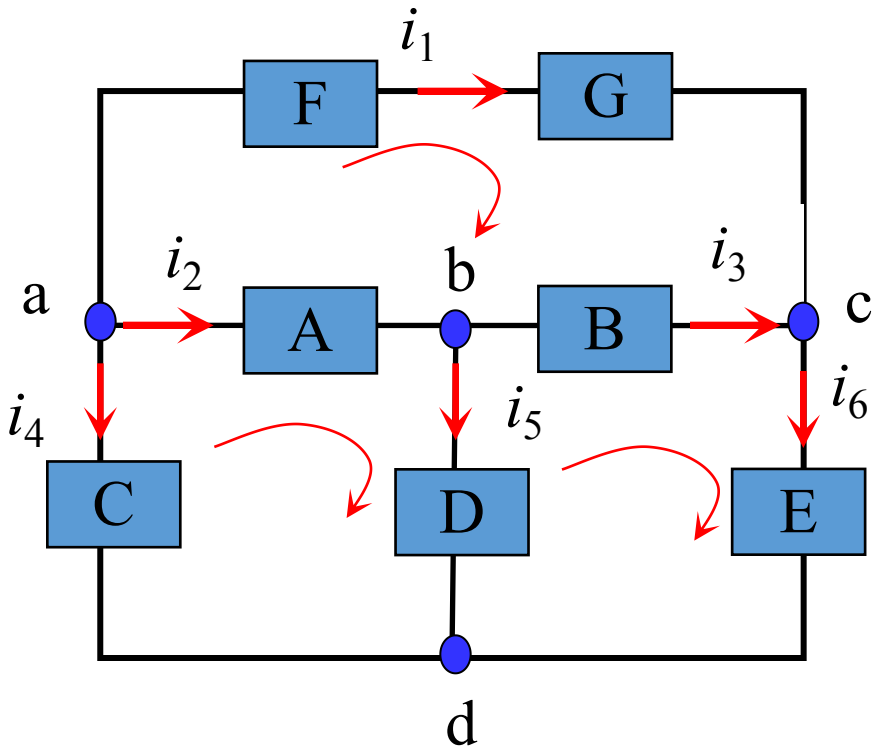
节点 $n=4$ $n-1$ 个独立的KCL方程

网孔 $m=3$ m 个独立的KVL方程

$m=b-n+1$ b 个方程

b 条支路 b 个伏安方程

$2b$ 个方程



二 → 支路电流法
(branch current method)

1 支路电流法

$2b$ 个变量

b 个支路电压　　$u = \mathrm{f}(i)$

b 个支路电流

$n-1$ 个独立的KCL方程

$b-n+1$ 个独立的KVL方程

b 个方程

$u = \mathrm{f}(i)$

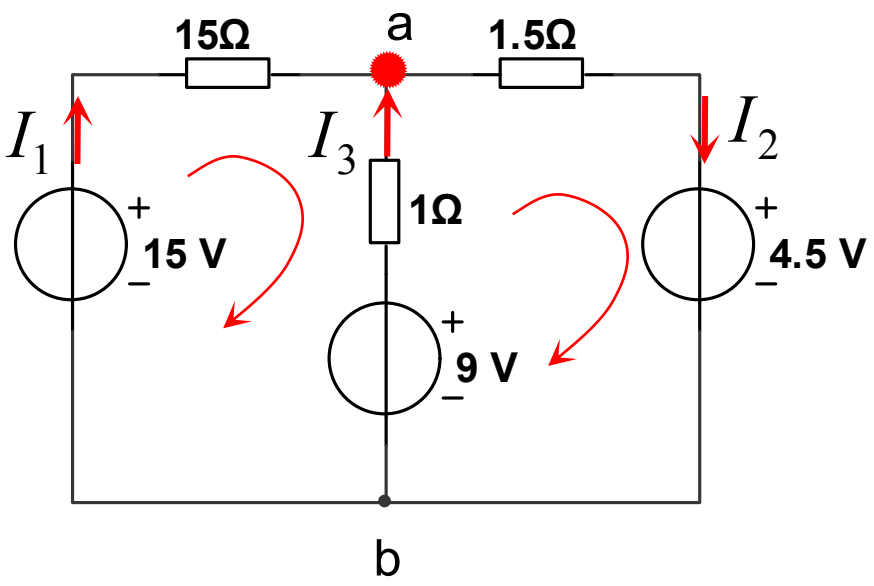
→ b 个支路电流 → b 个支路电压

例1 求图中电路各支路电流，并求 U_{ab} 及ab支路发出的功率。

解： $b=3 \quad n=2$

$$\begin{cases} I_1 - I_2 + I_3 = 0 \\ 15I_1 - I_3 = 15 - 9 \\ 1.5I_2 + I_3 = 9 - 4.5 \end{cases}$$

$$\begin{cases} I_1 = 0.5\ \text{A} \\ I_2 = 2\ \text{A} \\ I_3 = 1.5\ \text{A} \end{cases}$$



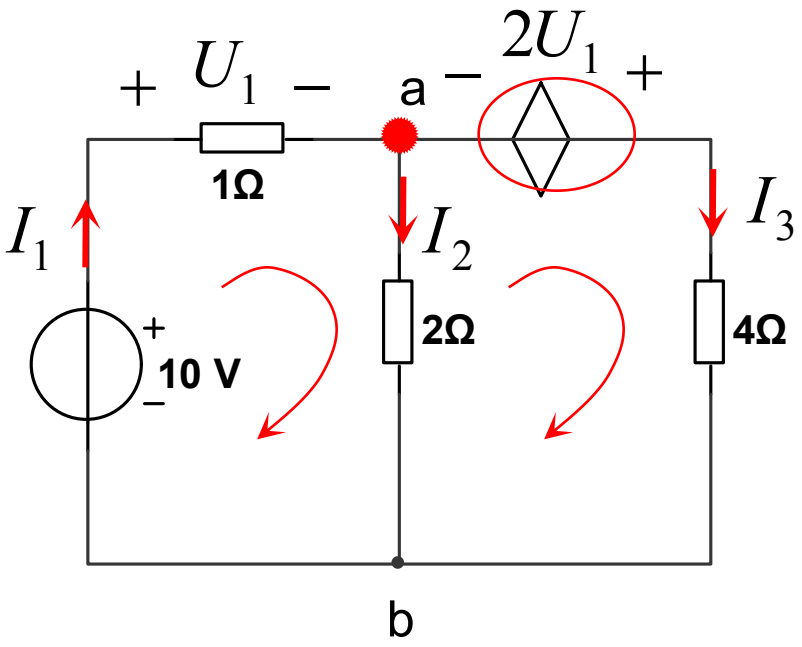
$$U_{ab} = 9 - I_3 = 9 - 1.5 = 7.5(\text{V})$$

$$P_{ab} = -U_{ab}I_3 = -7.5 \times 1.5 = -11.25(\text{W})$$

二 → 支路电流法
(branch current method)

2 支路电流法例题

例2 求图中电路各支路电流。



解： $b=3 \quad n=2$

把受控源当独立源看

$$\begin{cases} I_1 - I_2 - I_3 = 0 \\ I_1 + 2I_2 = 10 \\ 4I_3 - 2I_2 = 2U_1 \\ U_1 = I_1 \end{cases}$$

$$\begin{cases} I_1 = 6\ \text{A} \\ I_2 = 2\ \text{A} \\ I_3 = 4\ \text{A} \end{cases}$$

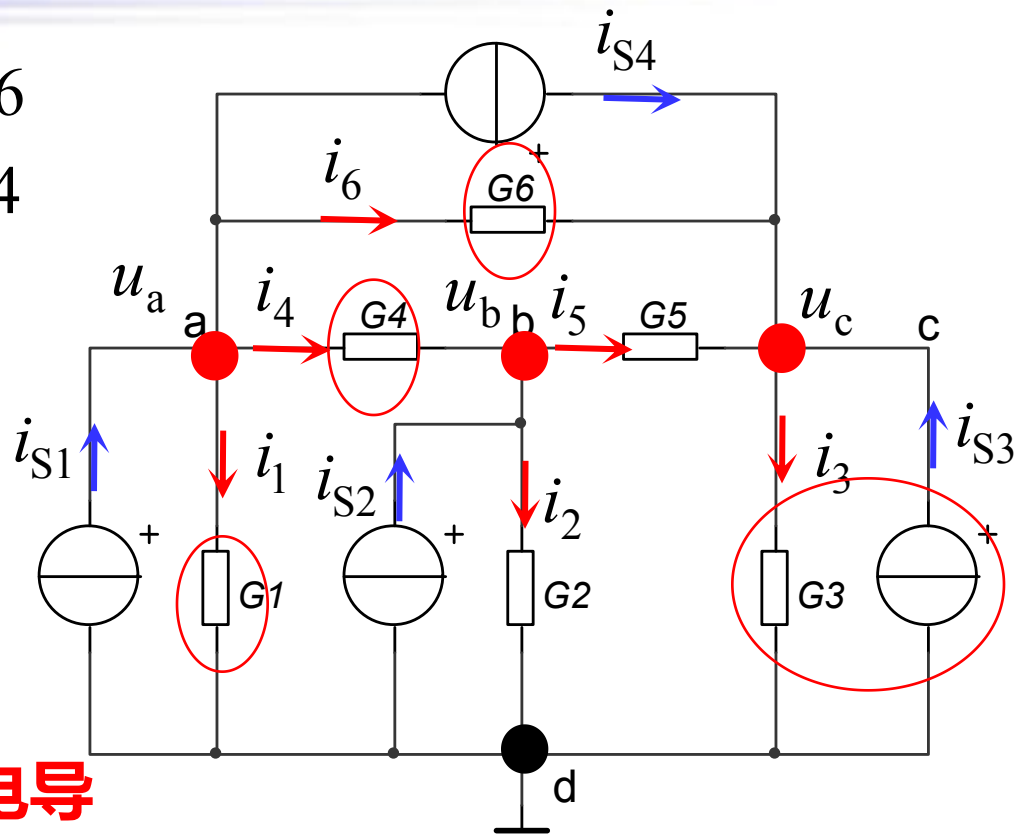
三 → 节点电压法
（node voltage method）

1 节点电压法

$$u_{ab} = u_a - u_b$$
$$u_{bc} = u_b - u_c$$
$$u_{ac} = u_a - u_c$$

$$i_1 = G_1 u_a$$
$$i_2 = G_2 u_b$$
$$i_3 = G_3 u_c$$
$$i_4 = G_4 u_{ab} = G_4(u_a - u_b)$$
$$i_5 = G_5 u_{bc} = G_5(u_b - u_c)$$
$$i_6 = G_6 u_{ac} = G_6(u_a - u_c)$$

$$b = 6$$
$$n = 4$$



$$\begin{cases} i_1 + i_4 + i_6 = i_{S1} - i_{S4} \\ i_2 - i_4 + i_5 = i_{S2} \\ i_3 - i_5 - i_6 = i_{S3} + i_{S4} \end{cases}$$

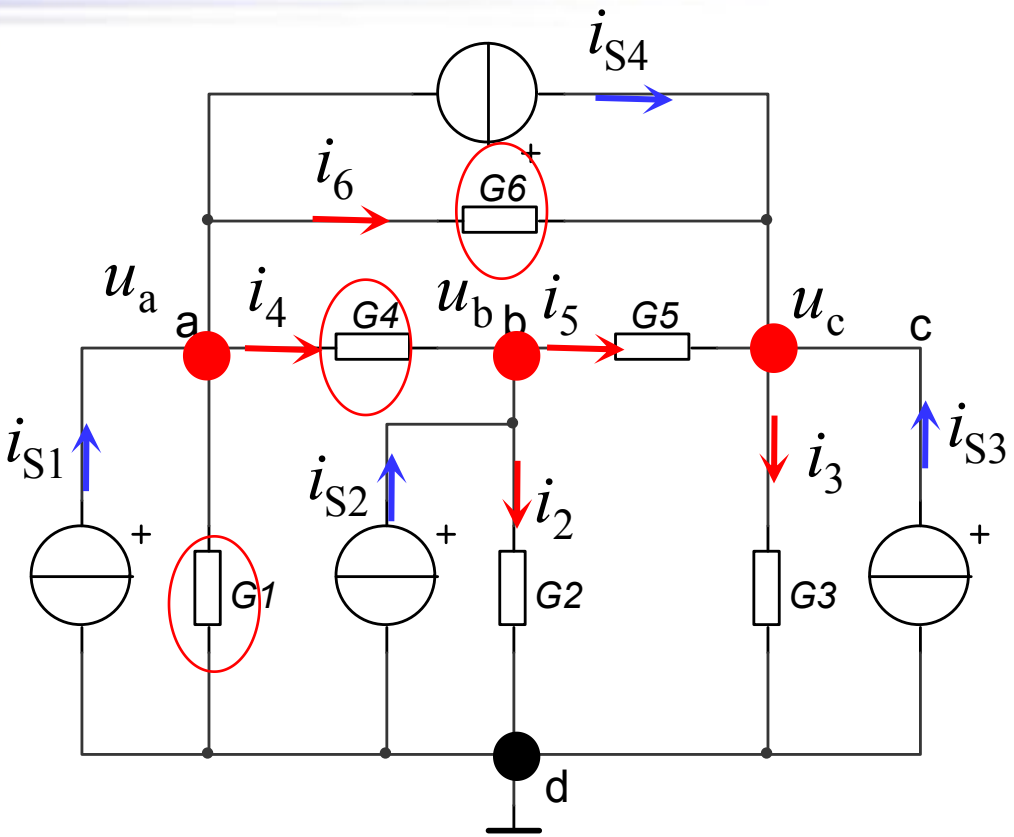
自电导　　互电导

$$\begin{cases} (G_1 + G_4 + G_6)u_a - (G_4 u_b + G_6 u_c) = i_{S1} - i_{S4} & \text{节点a} \\ (G_2 + G_4 + G_5)u_b - (G_4 u_a + G_5)u_c = i_{S2} & \text{节点b} \\ (G_3 + G_5 + G_6)u_c - (G_6 u_a + G_5 u_b) = i_{S3} + i_{S4} & \text{节点c} \end{cases}$$

自电导　　互电导

$$\begin{cases}(G_1+G_4+G_6)u_a-(G_4u_b+G_6u_c)=i_{S1}-i_{S4}\\(G_2+G_4+G_5)u_b-(G_4u_a+G_5u_c)=i_{S2}\\(G_3+G_5+G_6)u_c-(G_6u_a+G_5u_b)=i_{S3}+i_{S4}\end{cases}$$

本节点电压　　相邻节点电压



自电导✖本节点电压 ➖ ∑（互电导✖ 相邻节点电压）

＝ ∑（流入本节点电流源）

三 → 节点电压法
（node voltage method）

2 电流源串联电阻

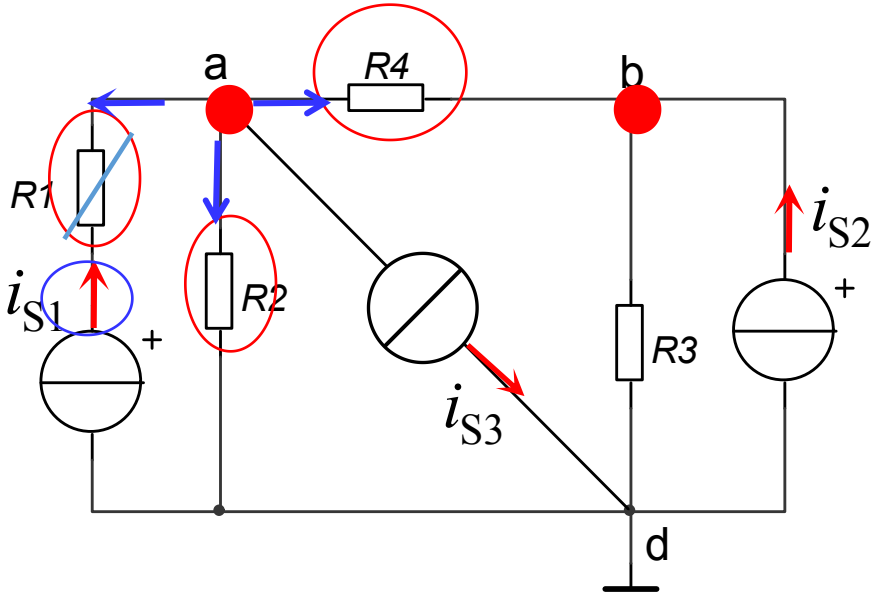
自电导✖本节点电压 — ∑（互电导✖ 相邻节点电压）

= ∑（流入本节点电流源）

∑电导上流出电流=∑电流源流入电流

$$(\frac{1}{R_2}+\frac{1}{R_4})u_a-\frac{1}{R_4}u_b=i_{S1}-i_{S3}$$

$$(\frac{1}{R_3}+\frac{1}{R_4})u_b-\frac{1}{R_4}u_a=i_{S2}$$



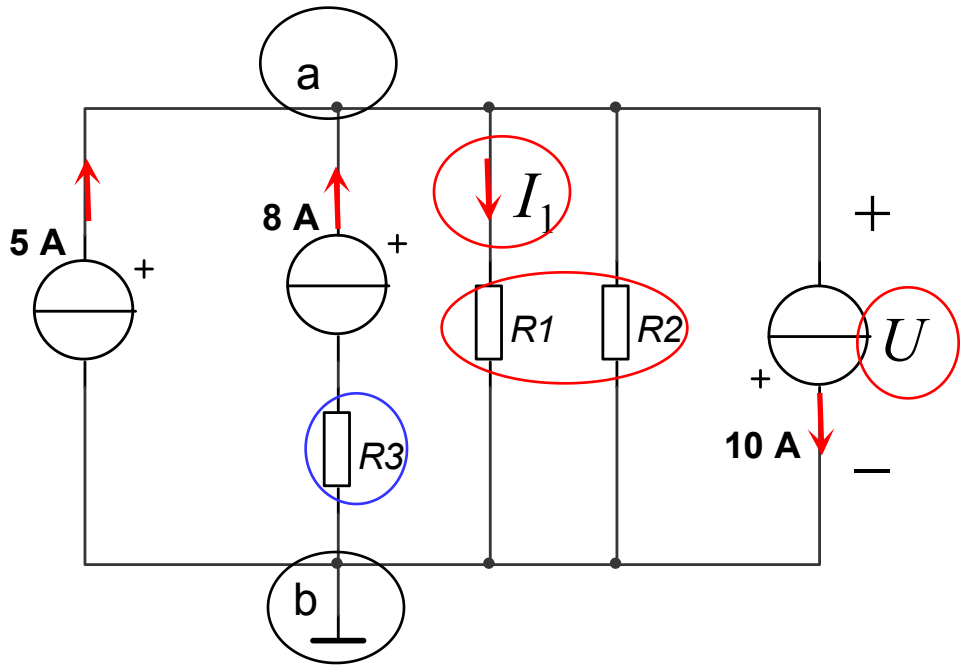
三 → 节点电压法
（node voltage method）

3 弥尔曼定理

$$(\sum G)\cdot u=\sum i_S$$

弥尔曼定理 $u=\dfrac{\sum i_S}{\sum G}$

例 已知 $R_1=5\Omega, R_2=10\Omega$, $R_3=15\Omega$, **求** I_1



解： $U=\dfrac{\sum I_S}{\sum G}=\dfrac{5+8-10}{\dfrac{1}{R_1}+\dfrac{1}{R_2}}=\dfrac{3}{\dfrac{1}{5}+\dfrac{1}{10}}=10(\mathrm{V})$

$$I_1=\frac{U}{R_1}=\frac{10}{5}=2(\mathrm{A})$$

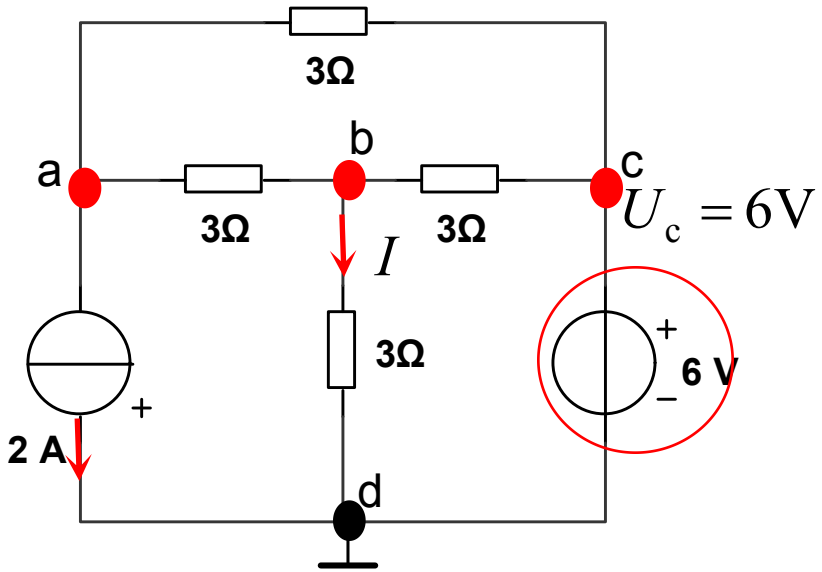
三 → 节点电压法
（node voltage method）

4 电路中含有电压源

例1 求电流 I

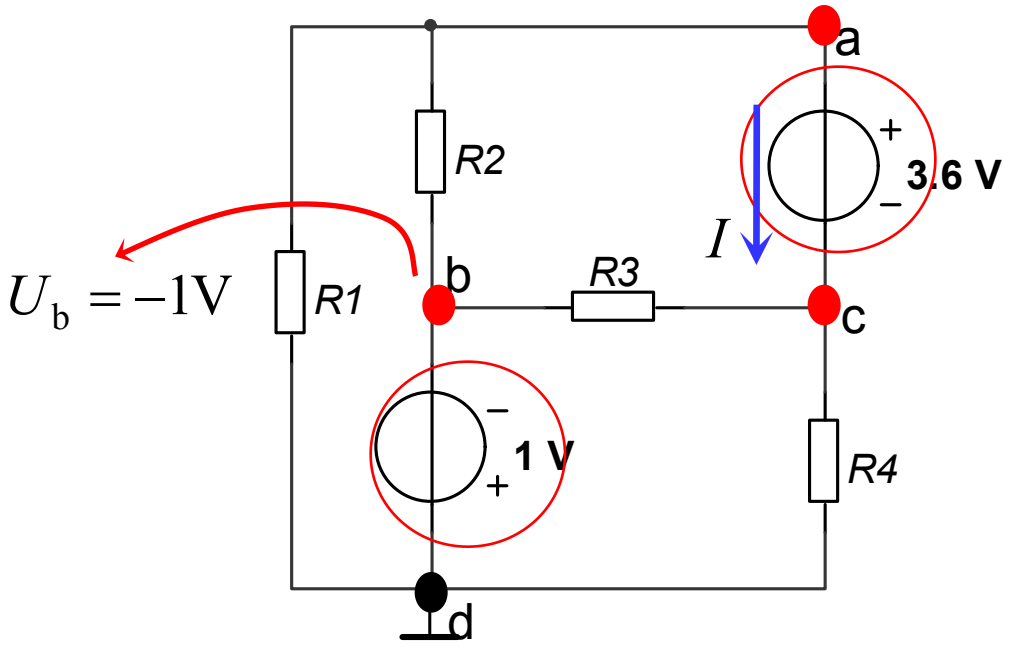
解：
$$\begin{cases}(\frac{1}{3}+\frac{1}{3})U_a-\frac{1}{3}U_b-\frac{1}{3}U_c=-2\\(\frac{1}{3}+\frac{1}{3}+\frac{1}{3})U_b-\frac{1}{3}U_a-\frac{1}{3}U_c=0\\U_c=6\text{V}\end{cases}$$

$$\begin{cases}U_a=1.2\text{V}\\U_b=2.4\text{V}\end{cases}\qquad I=\frac{U_b}{3}=0.8\text{A}$$



$$\begin{cases} \left(\frac{1}{R_1}+\frac{1}{R_2}\right)U_a-\frac{1}{R_2}U_b=-I \\ \left(\frac{1}{R_3}+\frac{1}{R_4}\right)U_c-\frac{1}{R_3}U_b=I \\ U_b=-1\text{V} \end{cases}$$

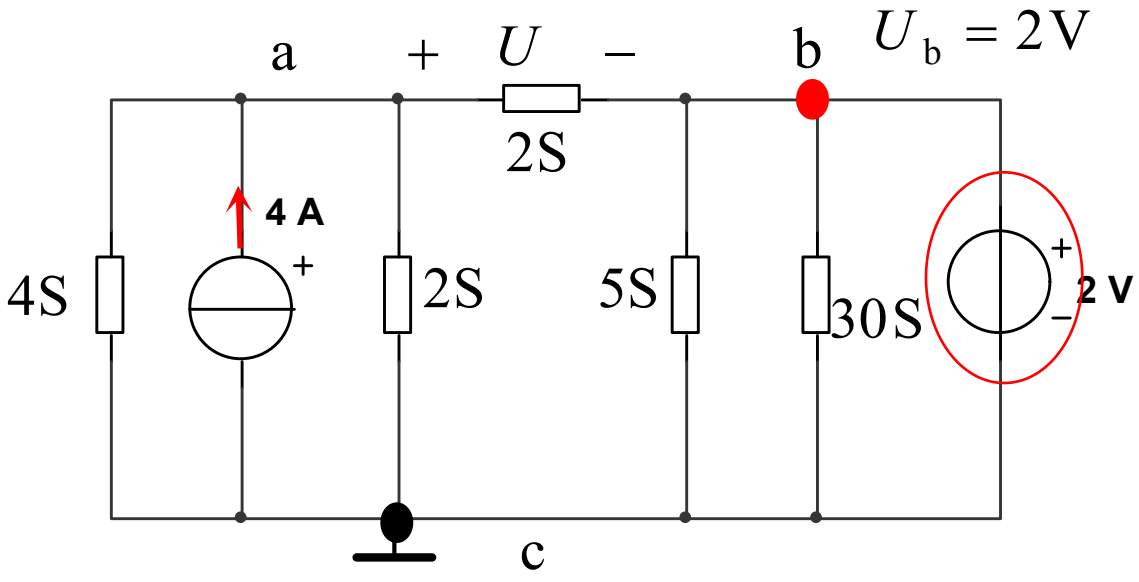
$$U_a-U_c=3.6$$



三 → 节点电压法
(node voltage method)

5 节点电压法例题

例1 电路如图，用节点电压法求电压*U*。

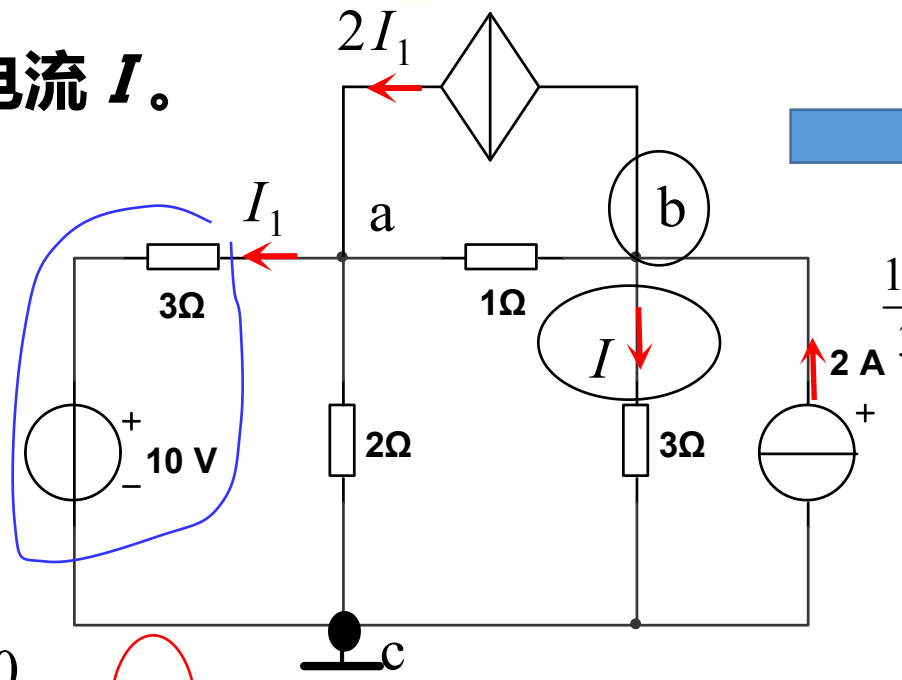


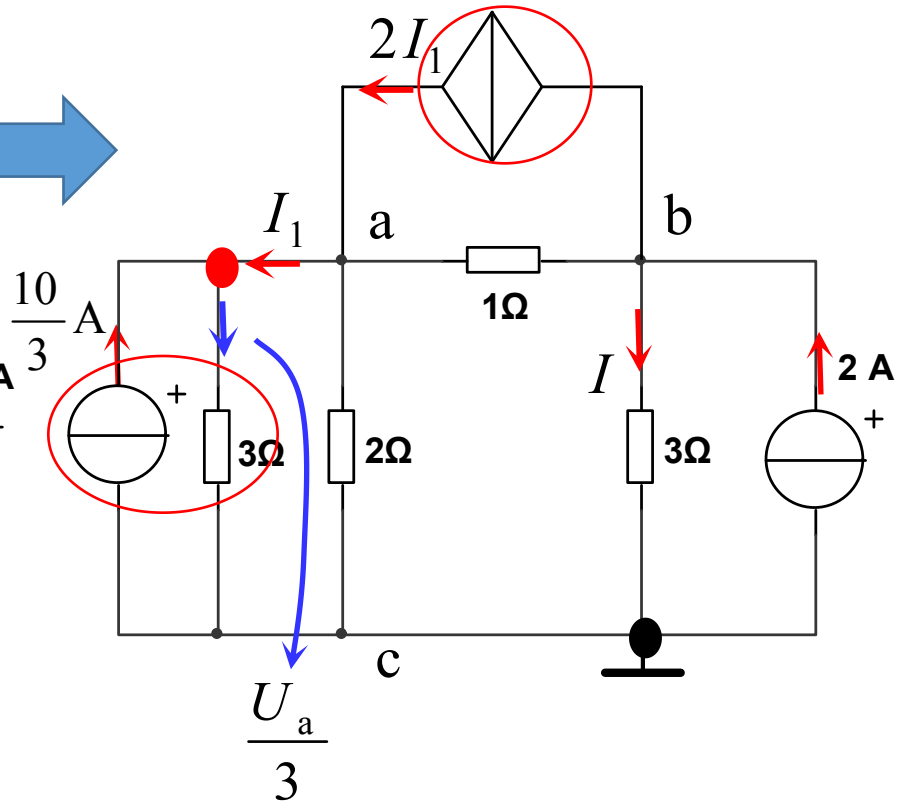
$$\begin{cases}(4+2+2)U_a - 2U_b = 4 \\ U_b = 2V\end{cases}$$

$$U_a = 1V$$

$$U = U_a - U_b = 1 - 2 = -1(V)$$

例2 电路如图，求电流 I。





$$\begin{cases} (\frac{1}{3}+\frac{1}{2}+1)U_a - U_b = \frac{10}{3} + 2I_1 \\ (\frac{1}{3}+1)U_b - U_a = 2 - 2I_1 \\ I_1 = \frac{U_a}{3} - \frac{10}{3} \end{cases}$$

$$U_b = 7.36\,\text{V}$$

$$I = \frac{U_b}{3} = 2.45\,(\text{A})$$

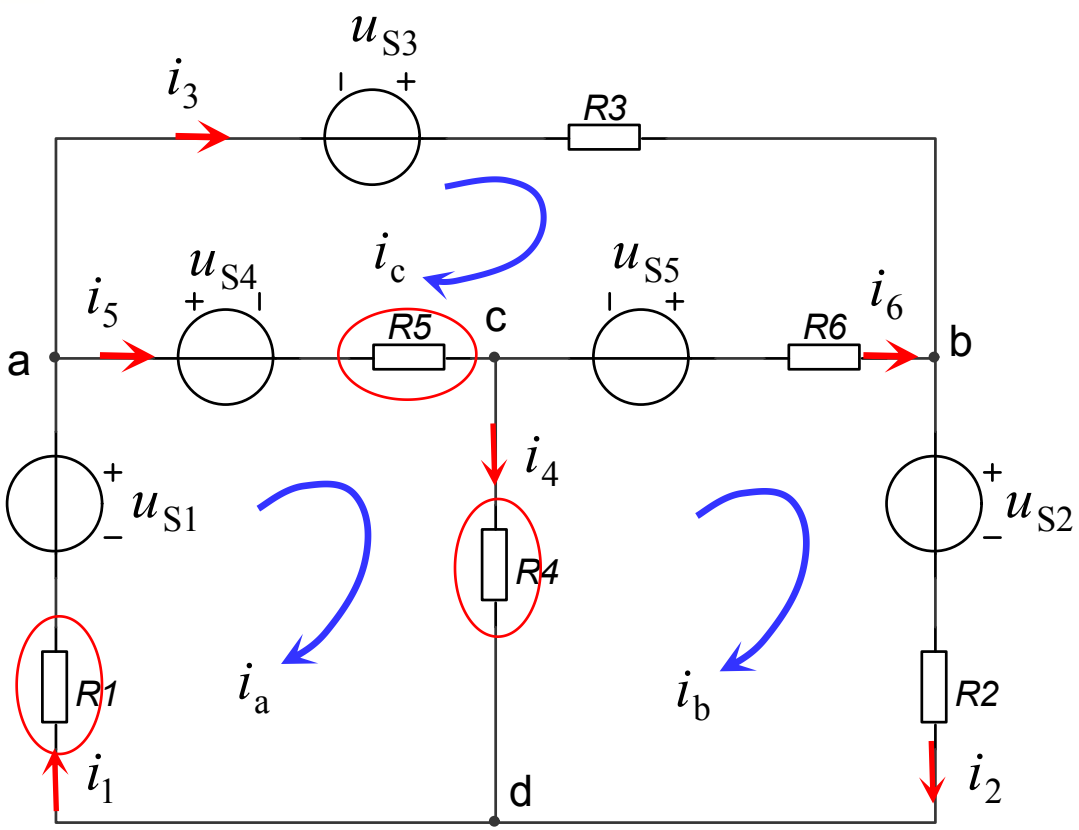
四 → 网孔电流法
（mesh current method）

1 网孔电流法

$$b = 6 \qquad n = 4 \qquad m = 3$$

$$\begin{cases} R_1 i_1 + R_4 i_4 + R_5 i_5 = u_{S1} - u_{S4} \\ R_2 i_2 - R_4 i_4 + R_6 i_6 = u_{S5} - u_{S2} \\ R_3 i_3 - R_5 i_5 - R_6 i_6 = u_{S3} + u_{S4} - u_{S5} \end{cases}$$

$$\begin{cases} i_1 = i_a \\ i_2 = i_b \\ i_3 = i_c \\ i_4 = i_a - i_b \\ i_5 = i_a - i_c \\ i_6 = i_b - i_c \end{cases}$$



自电阻　　互电阻

$$\begin{cases} (R_1 + R_4 + R_5) i_a - (R_4 i_b + R_5 i_c) = u_{S1} - u_{S4} \\ (R_2 + R_4 + R_6) i_b - (R_4 i_a + R_6 i_c) = u_{S5} - u_{S2} \\ (R_3 + R_5 + R_6) i_c - (R_5 i_a + R_6 i_b) = u_{S3} + u_{S4} - u_{S5} \end{cases}$$

自电阻 ✖ 本网孔电流 ± ∑（互电阻 ✖ 相邻网孔电流）

＝ ∑（本网孔中沿网孔方向电压源的电压升）

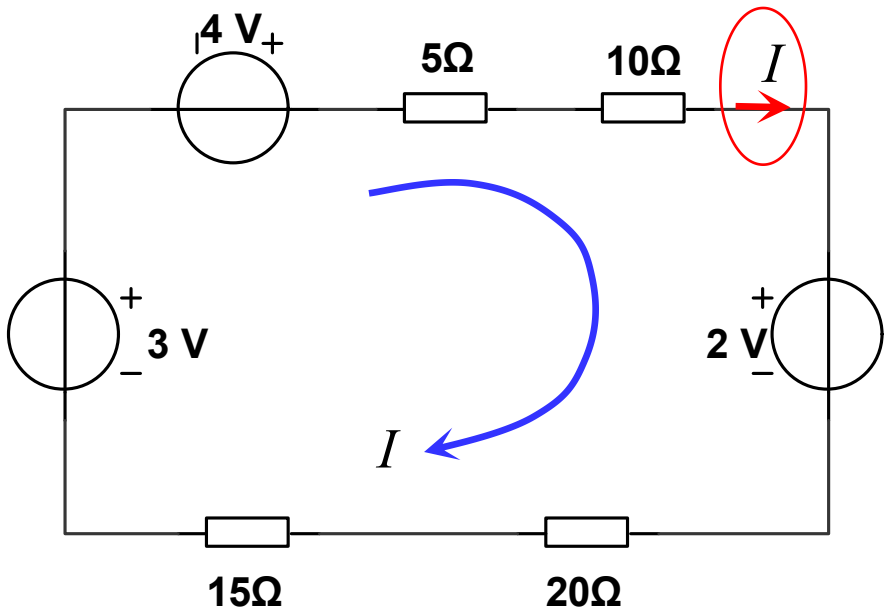
四 → 网孔电流法
（mesh current method）

2 全欧姆定律

$$(\sum R)\cdot I=\sum U$$

全欧姆定律 $$I=\frac{\sum U}{\sum R}$$

$$I=\frac{\sum U}{\sum R}=\frac{4+3-2}{5+10+20+15}=0.1(\mathrm{A})$$



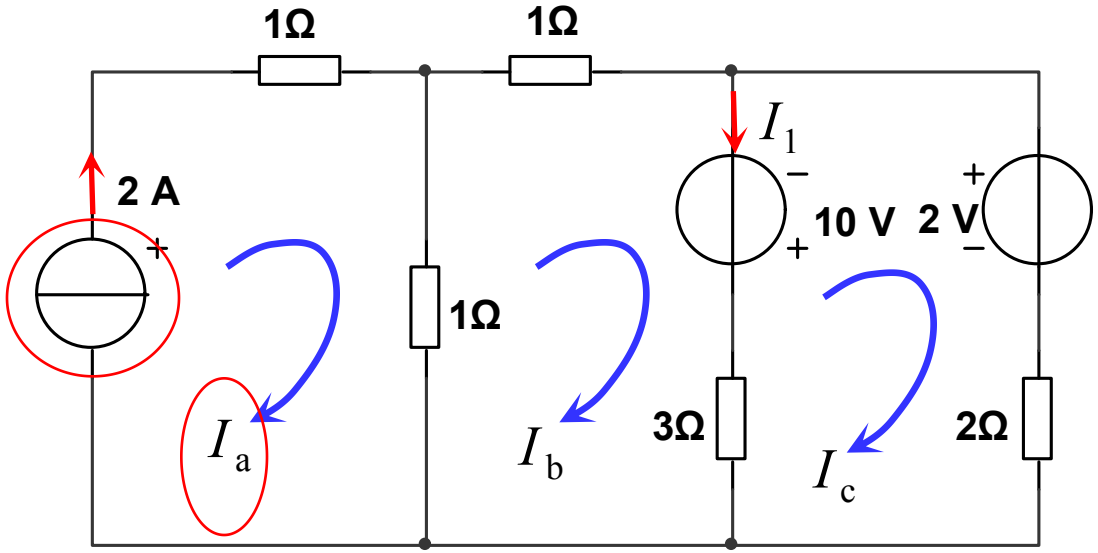
四 → 网孔电流法
（mesh current method）

3 电路中含有电流源

$$\begin{cases}(1+1+3)I_b - I_a - 3I_c = 10 \\ (2+3)I_c - 3I_b = -10-2 \\ I_a = 2\text{A}\end{cases}$$

$$\begin{cases}I_b = 1.5\text{A} \\ I_c = -1.5\text{A}\end{cases}$$

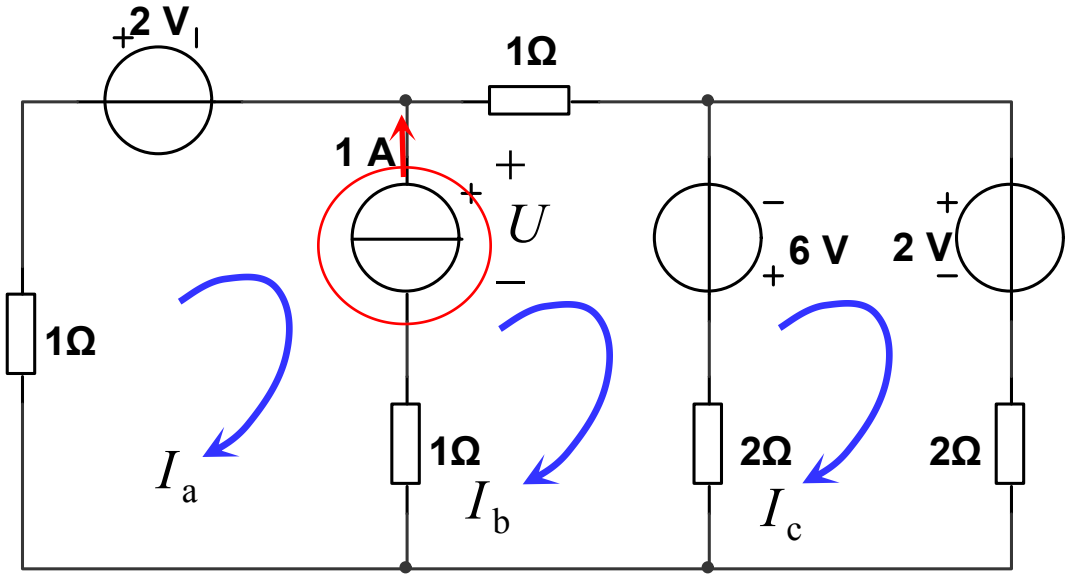
$$I_1 = I_b - I_c = 3(\text{A})$$



$$I_a = 2\text{A}$$

$$I_b - I_a = 1\text{A}$$

$$\begin{cases} (1+1)I_a - I_b = -2 - U \\ (1+1+2)I_b - I_a - 2I_c = U + 6 \\ (2+2)I_c - 2I_b = -6 - 2 \\ I_b - I_a = 1 \end{cases}$$

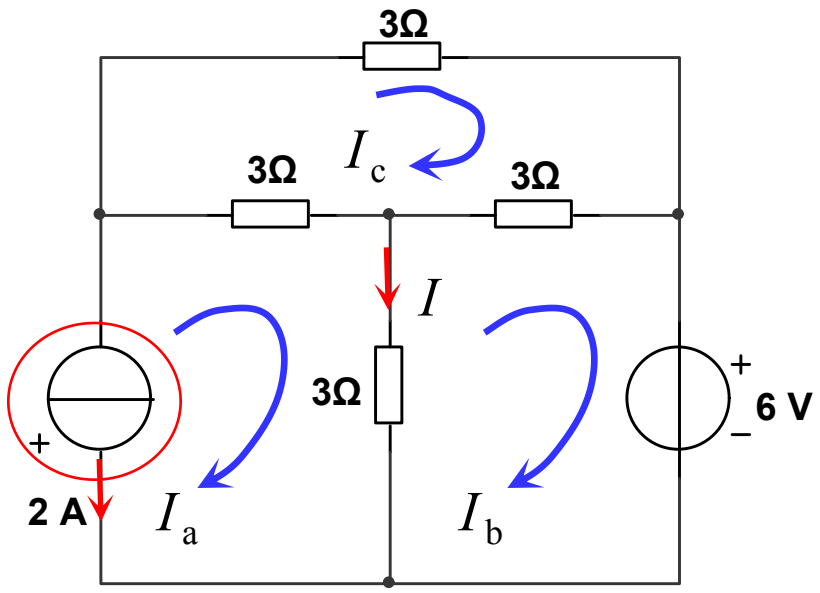


$$\begin{cases} I_a = -\dfrac{2}{3}\text{A} \\ I_b = \dfrac{1}{3}\text{A} \\ I_c = -\dfrac{11}{6}\text{A} \end{cases} \qquad U = -\frac{1}{3}\text{V}$$

四 → 网孔电流法
（mesh current method）

4 网孔电流法例题

例1 电路如图，求电流 I。



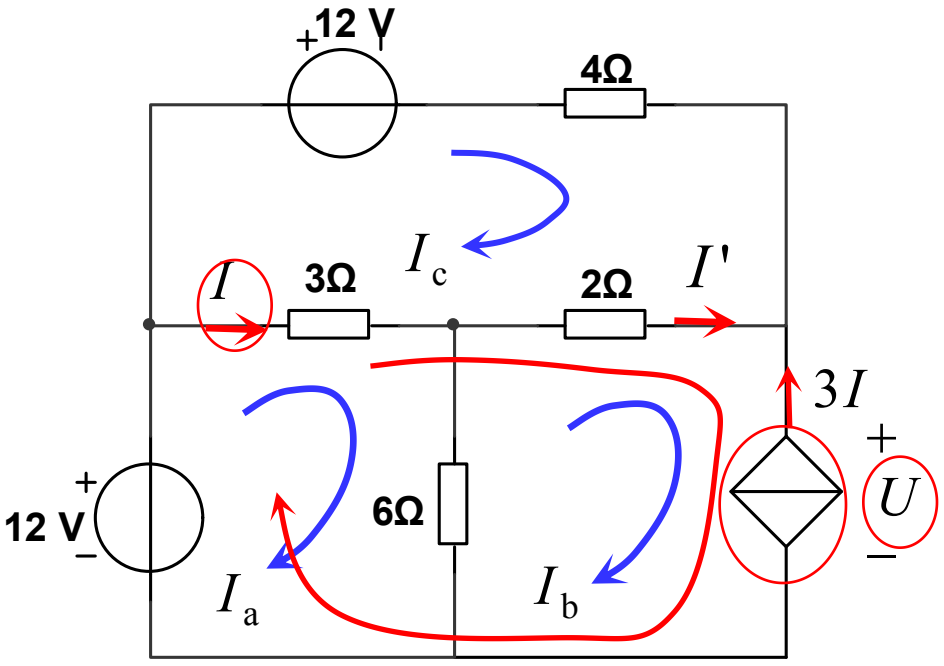
$$\begin{cases}(3+3)I_b-3I_a-3I_c=-6\\(3+3+3)I_c-3I_a-3I_b=0\\I_a=-2\end{cases}$$

$$I_a=-2\text{A}$$

$$I_b=-2.8\text{A}$$

$$I=I_a-I_b=0.8\text{A}$$

例2 电路如图，求受控源电压 U。



$$\begin{cases}(3+6)I_a-6I_b-3I_c=12\\(3+2+4)I_c-3I_a-2I_b=-12\\I_b=-3I\\I_a-I_c=I\end{cases}$$

$$I=-\frac{I_b}{3}=1\text{A}$$

$$I_b=-3I$$

$$\begin{cases}I_a=-1.5\text{A}\\I_b=-3\text{A}\\I_c=-2.5\text{A}\end{cases}$$

$$I'=I_b-I_c=-0.5\text{A}$$

$$U=12-3I-2I'=12-3\times1-2\times(-0.5)=10(\text{V})$$